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|  <b>Marwadi</b><br>University | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                 |
| <b>Subject: Digital Signal and Image Processing (01CT1513)</b>                                                 | <b>Aim:- Perform gray level operations images.</b>                                                                         |                                 |
| <b>Open handed problem -7</b>                                                                                  | <b>Date: 16-08-2026</b>                                                                                                    | <b>Enrolment No:92400133131</b> |

Github link :- <https://github.com/kaladiyafarhan9879/Digital-Signal-and-Image-Processing.git>

Aim :-

**Perform gray level operations images.**

Operations

- Read Digital Image
- Convert color image to grayscale image
- Image Negation
- Image Thresholding
- Image Gamma Correction

Gray Scale image

- Grayscale images are particularly useful when the color information is not relevant to the analysis or when the goal is to simplify the image data for computational purposes. They are a fundamental concept in image processing and computer vision.
- A grayscale image, often referred to as a black-and-white image, is an image in which each pixel is represented by a single intensity value, typically ranging from 0 (black) to 255 (white) in an 8-bit image. In a

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grayscale image, there is no color information; instead, it represents only the brightness or luminance of each pixel.

- The grayscale value at each pixel represents the intensity or brightness of that pixel. Higher values indicate brighter areas, while lower values indicate darker areas.

## Applications:

- Grayscale images are commonly used in various fields, including medical imaging (such as X-rays and MRI scans), computer vision, image processing, and document scanning. They are also used for artistic purposes in black-and-white photography.
- Grayscale images require less storage space compared to color images because they have only one channel instead of three. This can be advantageous in applications where storage capacity is limited.

```
Code;- import glob,os
import cv2,numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt

BASE_DIR=r"D:\sem-5\dsip\open handed 7"
IMAGE_DIR=BASE_DIR

files=[os.path.join(IMAGE_DIR,f"{i}.jpg") for i in range(1,6)]
if not all(os.path.exists(f) for f in files):
```

|                                                                                                                |                                                                                                                            |                                 |
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```

files=sorted(glob.glob(os.path.join(IMAGE_DIR,"*"))[:5])

images=[cv2.imread(f,cv2.IMREAD_GRAYSCALE) for f in files]
images=[img for img in images if img is not None]

def contrast_stretch(img):
    min_val,max_val=np.min(img),np.max(img)
    if max_val==min_val:
        return img
    return np.uint8((img.astype(np.float32)-min_val)*255.0/(max_val-min_val))

def histogram_equalization(img):
    return cv2.equalizeHist(img)

def gamma_correction(img,gamma=0.4):
    norm=img/255.0
    return np.uint8(np.power(norm,gamma)*255.0)

def log_transform(img):
    norm=img.astype(np.float32)
    c=255.0/np.log(1.0+np.max(norm))
    return np.uint8(c*np.log1p(norm))

def image_negative(img):
    return 255-img

methods=[
    ("Contrast Stretching",contrast_stretch),
    ("Histogram Equalization",histogram_equalization),
    ("Gamma Transformation ( $\gamma=0.4$ )",gamma_correction),
    ("Log Transformation",log_transform),
    ("Image Negative",image_negative)
]

enhanced_images=[]

for i,(name,func) in enumerate(methods):
    enh=func(images[i])

```

|                                                                                                                |                                                                                                                            |                                 |
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```

enhanced_images.append(enh)
cv2.imwrite(os.path.join(BASE_DIR,f"enhanced_image_{i+1}.png"),enh)

fig,axes=plt.subplots(5,2,figsize=(10,18))

for i in range(5):
    name,_=methods[i]
    axes[i,0].imshow(images[i],cmap="gray")
    axes[i,0].set_title(f"Image {i+1} - Original",fontsize=11,fontweight="bold")
    axes[i,0].axis("off")
    axes[i,1].imshow(enhanced_images[i],cmap="gray")
    axes[i,1].set_title(f"Image {i+1} - {name}",fontsize=11,fontweight="bold")
    axes[i,1].axis("off")

plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR,"comparison_results.png"),dpi=300,bbox_inches="tight")
plt.close()

fig_hist,axes_hist=plt.subplots(5,2,figsize=(12,16))

for i in range(5):
    name,_=methods[i]

axes_hist[i,0].hist(images[i].ravel(),bins=256,range=[0,256],color="gray",alpha=0.8)
    axes_hist[i,0].set_title(f"Image {i+1} Original Histogram",fontsize=10)
    axes_hist[i,0].set_xlim([0,256])

axes_hist[i,1].hist(enhanced_images[i].ravel(),bins=256,range=[0,256],color="indigo",alpha=0.8)
    axes_hist[i,1].set_title(f"Image {i+1} Enhanced Histogram ({name})",fontsize=10)
    axes_hist[i,1].set_xlim([0,256])

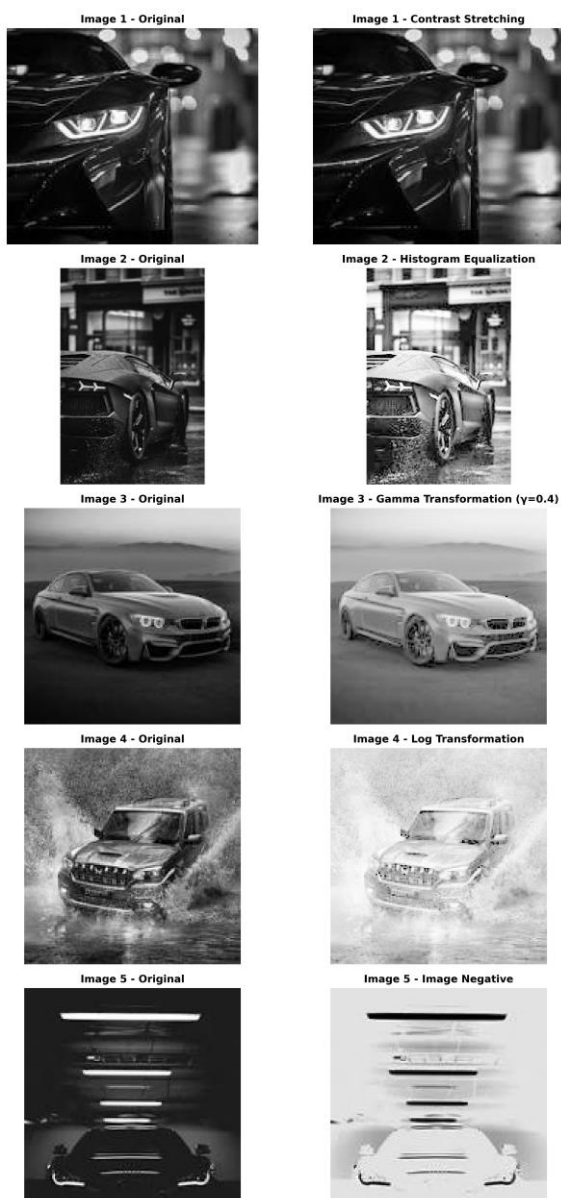
plt.tight_layout()

```

|                                                                                                             |                                                                                                                            |                                 |
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```
plt.savefig(os.path.join(BASE_DIR,"histogram_analysis.png"),dpi=300,bbox_inches="tight")
plt.close()
```

Output :-comperision image



|                                                                                                                |                                                                                                                            |                                 |
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Enhances image 1



Enhance image 2

|                                                                                                                |                                                                                                                            |                                 |
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Enhance image 3

|                                                                                                                |                                                                                                                            |                                 |
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Enhance image 4

|                                                                                                                |                                                                                                                            |                                 |
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Enahnse image 5

|                                                                                                                |                                                                                                                            |                                 |
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histogram\_analysis



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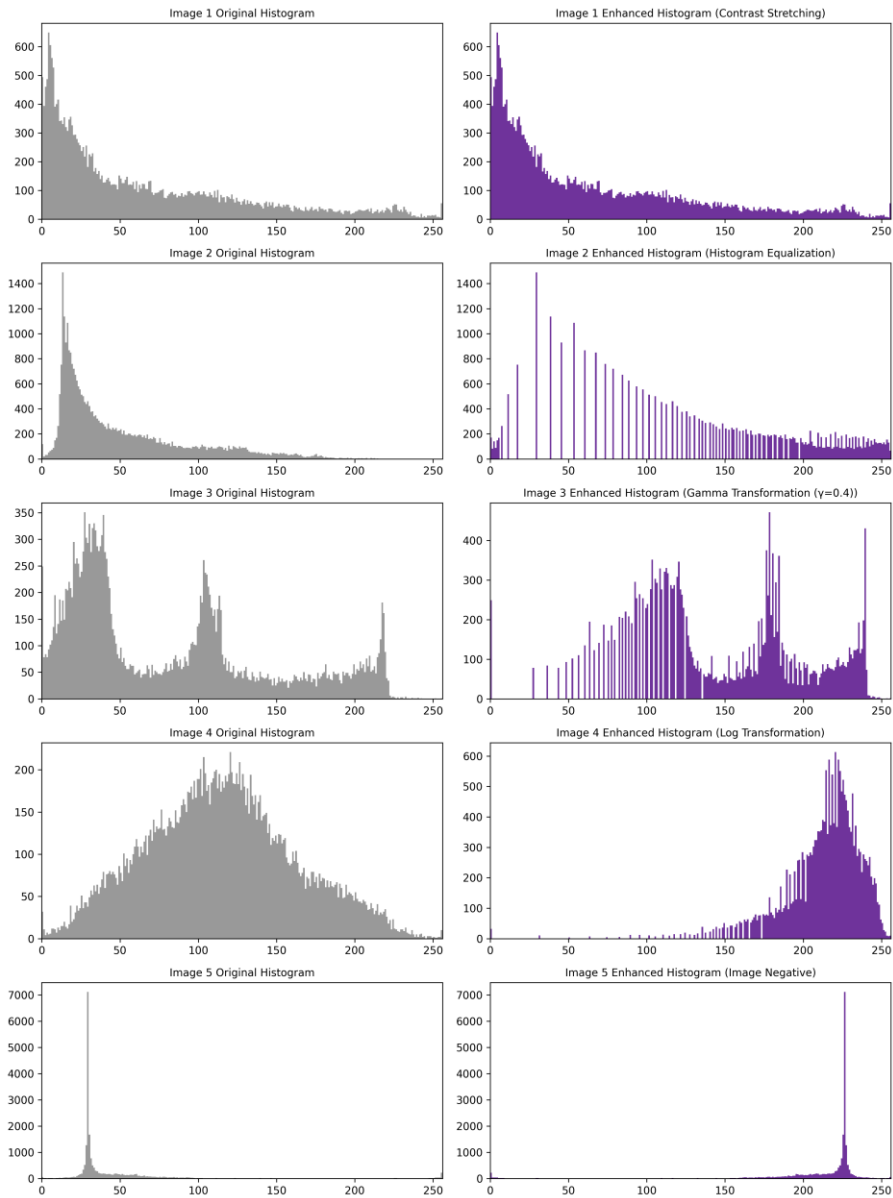
**Subject: Digital Signal and  
Image Processing  
(01CT1513)**

**Aim:- Perform gray level  
operations images.**

**Open handed problem -7**

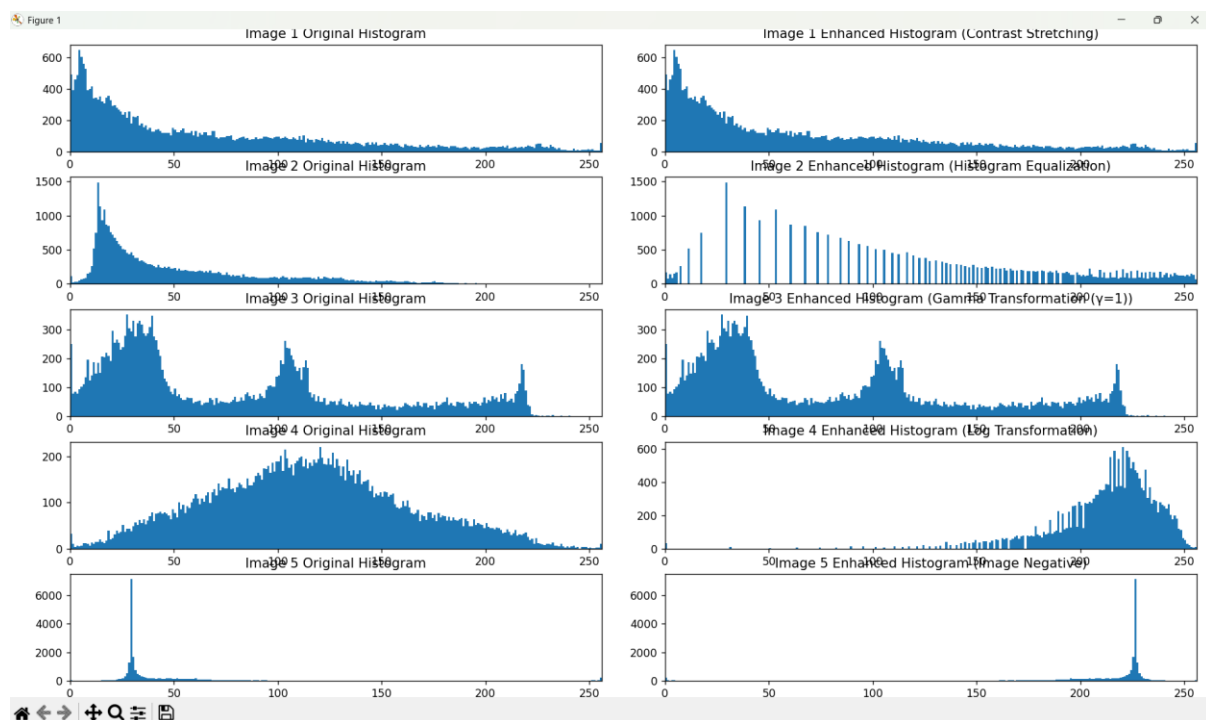
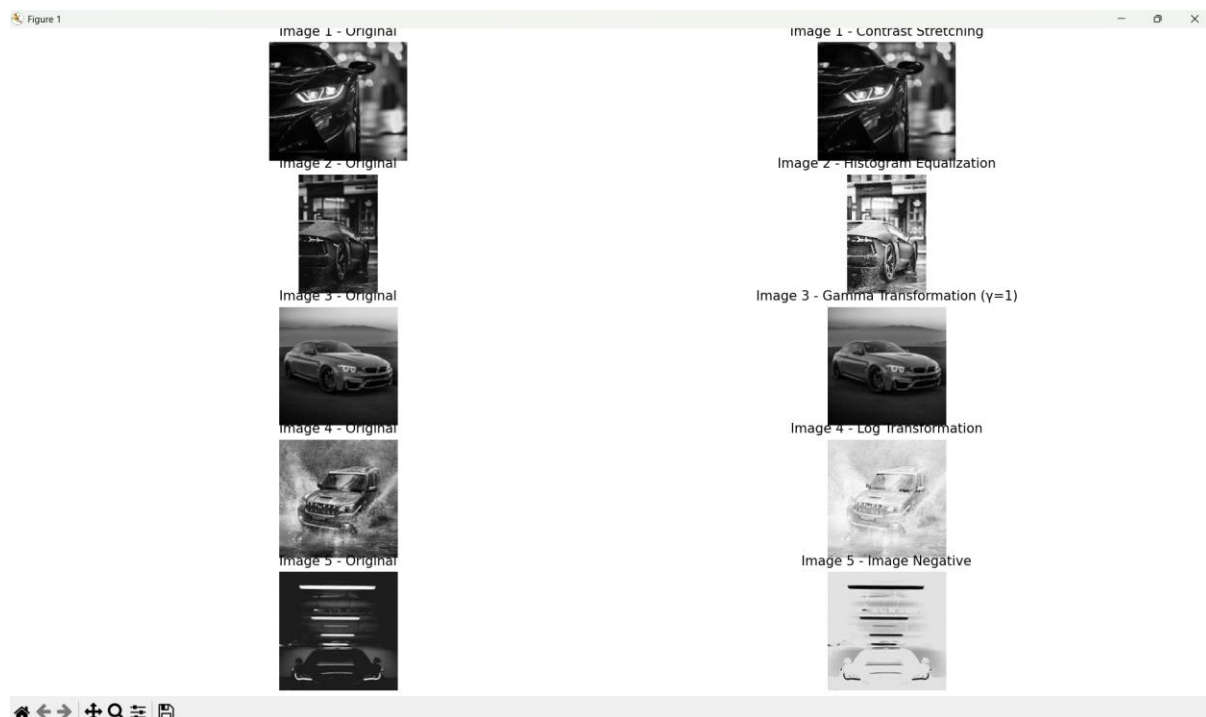
**Date: 16-08-2026**

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GAMMA 1

|                                                                                                             |                                                                                                                            |                                 |
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GAMMA = 2.2



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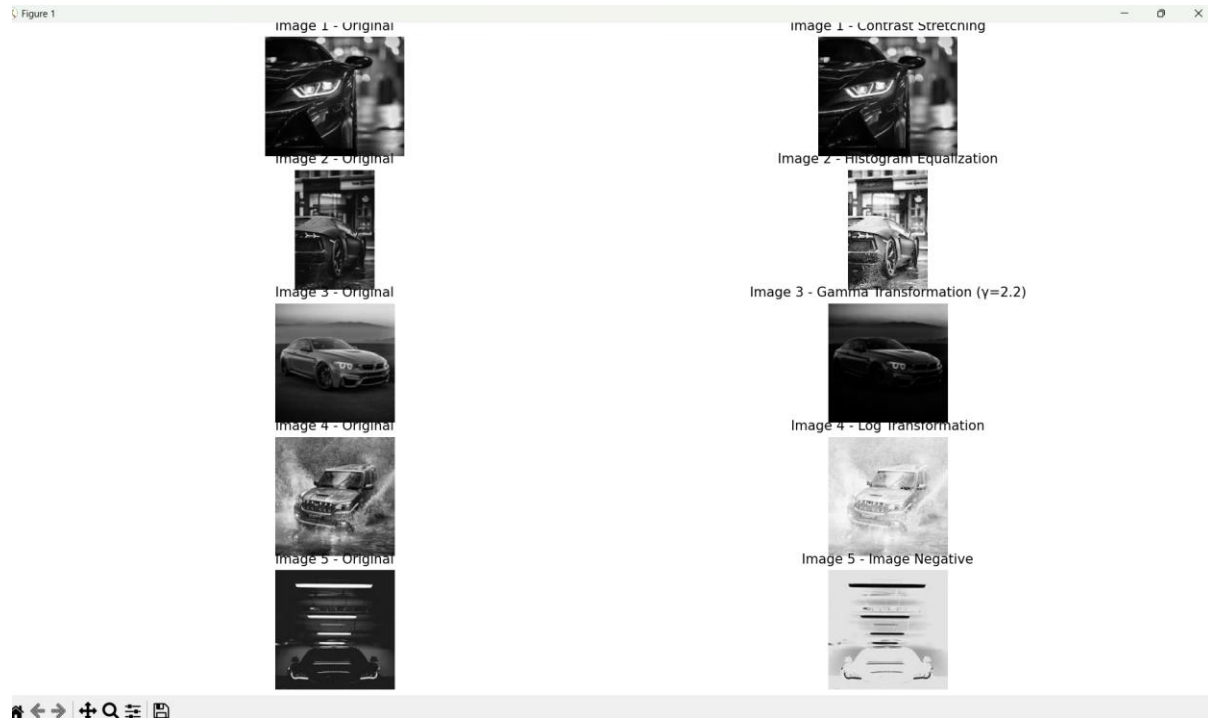
**Subject: Digital Signal and  
Image Processing  
(01CT1513)**

**Aim:- Perform gray level  
operations images.**

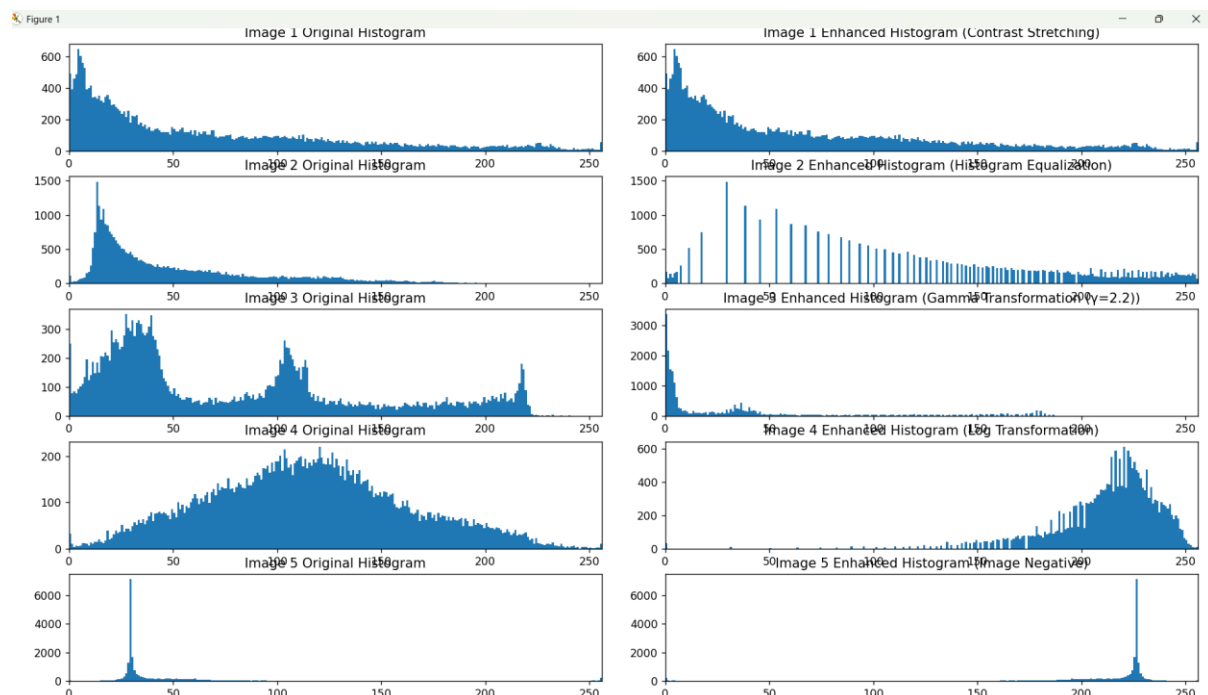
**Open handed problem -7**

**Date: 16-08-2026**

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The general transformation is represented as :  $s=T(r)$

Where:

?

$r$  = original pixel intensity

?

$s$  = transformed pixel intensity

?

$T$  = transformation function

2. Contrast Stretching Contrast stretching improves images with low contrast by expanding the range of pixel intensities. It makes dark areas darker and bright areas brighter, improving overall image clarity.

|                                                                                                                |                                                                                                                            |                                 |
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3. Histogram Equalization Histogram equalization improves image contrast by redistributing the intensity values. It helps reveal details that may not be clearly visible in the original image.
  
4. Gamma Transformation Gamma transformation changes the brightness of an image using a nonlinear intensity adjustment. A gamma value less than 1 mainly brightens dark regions and reveals hidden details.
  
5. Log Transformation Log transformation enhances details in dark regions of an image. It increases the visibility of low-intensity pixels while reducing the effect of very bright pixels.
  
6. Image Negative Image negative reverses the brightness of an image. Dark regions become bright and bright regions become dark, which can make certain details easier to observe.

|                                                                                                                |                                                                                                                            |                                 |
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7. Histogram A histogram shows how pixel intensity values are distributed in an image. Comparing original and enhanced histograms helps analyze the effect of each enhancement technique.

## Conclusion

1. Different image enhancement techniques were successfully implemented using Python and OpenCV.
2. Contrast stretching improved the intensity range and enhanced image details.
3. Histogram equalization improved the overall contrast of the images.
4. Gamma transformation adjusted the brightness of the images effectively.
5. Log transformation enhanced details, especially in darker regions.
6. Image negative produced the inverse intensity of the original image.
7. Original and enhanced images were displayed side by side for comparison.
8. Histograms were plotted to analyze the distribution of pixel intensities.

|                                                                                                                |                                                                                                                            |                                 |
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9. Each enhancement technique produced different visual results.
10. The experiment showed that image enhancement improves image quality and visibility.
11. OpenCV provided efficient functions for image processing operations.
12. Matplotlib was used to visualize images and their histograms.
13. The experiment provided practical understanding of spatial-domain image enhancement techniques.